

NATURAL LANGUAGE PROCESSING



Morphological Analysis

How machines break words into meaning — the foundational phase of the NLP pipeline that studies word structure, form, and formation.

NLP • Computational Linguistics • Language Structure

What We'll Cover

01 NLP pipeline & where morphology fits

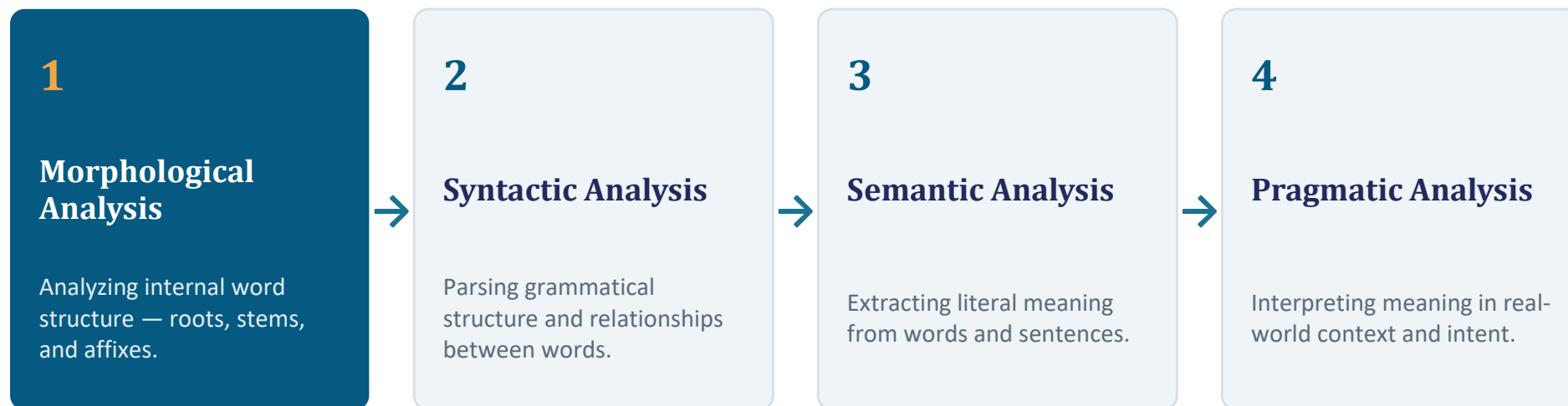
02 Foundations: morphemes, roots, stems, affixes

03 Word formation processes

04 Ambiguity, irregularity & challenges

05 Applications & tools

Where Morphological Analysis Fits



Why Start With Morphology?

Every downstream NLP task depends on knowing what a word actually is. Before a system can tag part-of-speech, parse a sentence, or translate a phrase, it must reduce surface word forms to something normalized and analyzable.

Morphological analysis supplies that foundation: it identifies a word's root, its grammatical category, and the meaning contributed by each piece attached to it — turning an unbounded vocabulary of surface forms into a manageable, structured representation.

FEEDS INTO

- Part-of-speech tagging
- Syntactic parsing
- Machine translation
- Information retrieval
- Speech synthesis

Morphology

linguistics · noun

The branch of linguistics concerned with the internal structure of words — how words are formed from smaller meaningful units, and how those units combine to signal grammatical and semantic information.

EXAMPLES

cats

cat + -s (plural)

unhappiness

un- + happy + -ness

walked

walk + -ed (past tense)

Morphological Analysis

NLP · computational task

The computational process of identifying the morphemes that make up a word, and recovering its root form together with the grammatical features (tense, number, case, person) encoded by its affixes.

EXAMPLES

"running" → run + VERB + PROG

root + POS + grammatical feature

"studies" → study + VERB + 3sg

root + POS + agreement feature

Goals of Morphological Analysis

1

Identify the root/lemma — recover the dictionary base form underlying any inflected surface word.

2

Extract grammatical features — detect tense, number, gender, case, person, and aspect encoded in affixes.

3

Segment word structure — split a word into its constituent morphemes with clear boundaries.

Why Morphological Analysis Matters



Vocabulary Reduction

Collapses thousands of surface forms into a manageable set of root lemmas for modeling.



Morphologically Rich Languages

Essential for Turkish, Finnish, Hindi, Arabic — where a single word carries a full phrase's meaning.



Agreement & Grammar

Recovers grammatical relationships needed for accurate parsing and generation.

The Morpheme: Smallest Unit of Meaning



A morpheme is the smallest meaningful unit in a language — it cannot be divided further without losing or destroying its meaning. Words may consist of a single morpheme ("dog") or several combined morphemes ("un-believ-able"), each contributing its own piece of meaning or grammatical function.

Free vs. Bound Morphemes

Free Morphemes

- Stand alone as complete words
- Carry independent meaning
- Examples: book, run, happy, slow
- Can be content words (noun, verb) or function words (the, and)

Bound Morphemes

- Cannot stand alone — must attach to a base
- Include all affixes: prefixes & suffixes
- Examples: -s, -ed, -ing, un-, re-, -ness
- Always modify or extend a free morpheme's meaning

Every word is built from at least one free morpheme, optionally combined with one or more bound morphemes.

Types of Affixes



Prefix

Attaches before the root. English: un-, re-, dis-, pre-.



Suffix

Attaches after the root. English: -s, -ed, -ing, -ness.



Infix

Inserted inside the root itself. Rare in English; common in Tagalog, Khmer.



Circumfix

Wraps around the root as two parts. German: ge-...-t (gemacht).



Interfix

A linking element joining two roots in a compound (speed-o-meter).



Transfix

A discontinuous pattern interleaved with root consonants — Arabic k-t-b templates.

Prefix and Suffix in Action



Prefixes and suffixes are the most common affix types in English. Here, **dis-** reverses the meaning of the root, while **-ment** converts the verb "agree" into a noun — two affixes doing two very different jobs on the same root.

Word Formation Processes

A

Affixation

Adding prefixes/suffixes to a root: happy → unhappy.

C

Compounding

Joining two free morphemes: tooth + brush → toothbrush.

R

Reduplication

Repeating all/part of a root: bye-bye, wishy-washy.

B

Blending

Merging parts of two words: smoke + fog → smog.

X

Clipping

Shortening a longer word: advertisement → ad.

Inflectional Morphology

grammar-marking process

Modifies a word to express grammatical features — tense, number, case, person, gender, or aspect — without changing its part of speech or creating a new dictionary entry.

EXAMPLES

walk → walked

present → past tense

cat → cats

singular → plural

big → bigger

positive → comparative

he → him

nominative → accusative case

Derivational Morphology

word-creation process

Creates a new word — often in a different part of speech or with a distinctly new meaning — by attaching a derivational affix to a root or stem.

EXAMPLES

happy → happiness

adjective → noun

act → actor

verb → noun (agent)

nation → national

noun → adjective

modern → modernize

adjective → verb

Compounding: Two Roots, One Word



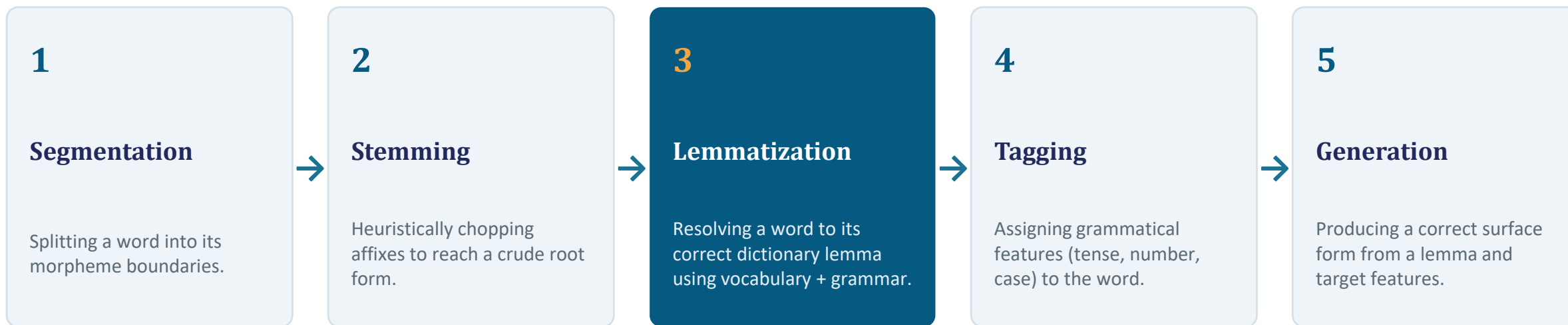
Compounding combines two or more independent free morphemes into a single new word. English compounds may be written solid (toothbrush), hyphenated (mother-in-law), or open (ice cream) — a surface inconsistency that complicates automatic tokenization and analysis.

02

Morphological Analysis in Practice

The core computational tasks and techniques

Morphological Analysis Task Family



Stemming

heuristic normalization

A rule-based process that strips affixes from a word to produce a "stem" — a crude approximation of the root that may not itself be a valid word, traded off against speed and simplicity.

EXAMPLES

studies → studi

not a real word, but a consistent stem

running → run

correct by coincidence of simple rule

caring → car

over-stemming: wrong result

The Porter Stemmer Algorithm

1

Step 1 — strip plural and past-tense/gerund suffixes (-sses, -ies, -s; -eed, -ed, -ing).

2

Step 2 — convert derivational suffixes to a simpler form (-ational → -ate, -iveness → -ive).

3

Step 3 — further reduce derivational suffixes (-icate → -ic, -ful, -ness removed).

4

Step 4 — strip common Latinate/Greek suffixes when the stem is long enough (-al, -ance, -ence, -ic).

5

Steps 5a/5b — clean up trailing "e" and duplicate consonants for a tidy final stem.

Porter Stemmer — Walkthrough



national

step 3: -ization → -ize? no, drop -ation

nation

step 4: strip -al

"nationalization" → "national" → "nation" over successive rule passes. Each Porter step applies only when the resulting stem meets a minimum length/measure condition — preventing over-aggressive stripping on short words.

Over-Stemming vs. Under-Stemming

Over-Stemming

- Strips too much; unrelated words collapse together
- Example: "university" and "universe" → "univers"
- Hurts precision — false matches in search/IR

Under-Stemming

- Strips too little; related words stay distinct
- Example: "data" and "datum" not unified
- Hurts recall — related documents missed

Stemmer design is a constant trade-off between these two failure modes.

Lemmatization

dictionary-based normalization

Reduces a word to its canonical dictionary form (lemma) using vocabulary lookup and morphological/grammatical analysis — always yielding a valid word, unlike stemming.

EXAMPLES

better → good

irregular comparative resolved via lexicon

was → be

requires POS context to disambiguate

geese → goose

irregular plural, not affix-strippable

Choosing the Right Tool

Stemming

- Rule-based, fast, no dictionary needed
- Output may not be a real word
- Ignores part-of-speech context
- Good fit: search engines, IR at scale

Lemmatization

- Dictionary + grammar aware, more accurate
- Output is always a valid lemma
- Uses POS tags to disambiguate
- Good fit: chatbots, MT, downstream NLU

The right choice depends on whether the application needs speed (stemming) or linguistic accuracy (lemmatization).



03

Ambiguity & Challenges

Where morphological analysis gets hard

Morphological Ambiguity

see + PAST

reading 1: verb

saw + NOUN

reading 2: tool

The same surface form can map to more than one valid morphological analysis. "Saw" could be the irregular past tense of "see", or the noun for a cutting tool — resolving the ambiguity requires sentence context, not the word form alone.

Open Challenges in Morphological Analysis

1

Irregular forms — go/went, good/better, child/children resist affix-based rules entirely and need lexical lookup.

2

Productivity vs. exceptions — rules that mostly work still have exceptions that must be special-cased.

3

Out-of-vocabulary words — new coinages, loanwords, and typos fall outside any fixed lexicon.

4

Resource scarcity — most robust analyzers exist only for a handful of well-resourced languages.

Where Morphological Analysis Is Used

M Machine Translation

Aligning inflectional systems that differ between source and target languages.

I Information Retrieval

Matching queries and documents that share a root but differ in surface form.

S Spell Checking

Validating word forms against known roots and legal affix combinations.

T Text-to-Speech

Determining correct pronunciation and stress from morphological structure.

Q Question Answering

Normalizing question and passage words to match despite inflection.

C Text Classification

Reducing feature space by collapsing inflectional variants of the same root.

Libraries & Toolkits

Tool	Approach	Typical Use
NLTK (Porter/Snowball/Lancaster)	Rule-based stemming	Teaching, prototyping, English IR
spaCy	Statistical/neural lemmatizer	Production NLP pipelines
HFST / Xerox FST	Finite-state transducers	Rule-based analyzers for many languages
Morfessor	Unsupervised segmentation	Low-resource, agglutinative languages
Stanford CoreNLP / Stanza	Statistical + neural	Multilingual tagging & lemmatization
SentencePiece / BPE	Subword tokenization	Preprocessing for large language models



Morphology turns surface text into structured meaning.

Every downstream NLP task — tagging, parsing, translation, retrieval — inherits the quality of this first analytical step

The better the analysis, the more every later stage can rely on it.